SQL Data Analysis Project

--1. Total Number of Patients

-- Write an SQL query to find the total number of patients across all hospitals.

```
SELECT

SUM(Patients_Count)

AS Total_Patients

FROM hospital_data;
```

--2. Average Number of Doctors per Hospital

-- Retrieve the average count of doctors available in each hospital.

```
SELECT

Hospital_Name,

CAST(AVG(CAST(Doctors_Count AS FLOAT)) AS DECIMAL(10,2))
AS Average_Doctor

FROM hospital_data

GROUP BY Hospital_Name;
```

--3. Top 3 Departments with the Highest Number of Patients

-- Find the top 3 hospital departments that have the highest number of patients.

```
SELECT
```

```
TOP 3
Department,
SUM(Patients_Count) AS Total_Patients
FROM hospital_data
GROUP BY Department
ORDER BY Total_Patients DESC;
```

--4. Hospital with the Maximum Medical Expenses

-- Identify the hospital that recorded the highest medical expenses.

```
SELECT
TOP 1
Hospital_Name,
SUM(Medical_Expenses) AS Total_Expenses
FROM hospital_data
GROUP BY Hospital_Name
ORDER BY Total_Expenses DESC;
```

--5. Daily Average Medical Expenses

-- Calculate the average medical expenses per day for each hospital.

```
SELECT
```

Hospital_Name,

SUM(Medical_Expenses) / SUM(DATEDIFF(day,Admission_Date, Discharge_Date)) AS Daily_Average_Medical_Expenses

FROM

hospital_data

GROUP BY

Hospital_Name;

--6. Longest Hospital Stay

-- Find the patient with the longest stay by calculating the difference between Discharge Date and Admission Date.

SELECT

TOP 1

Patients_Count,

DATEDIFF(day,Admission_Date,Discharge_Date) AS Total_Stay_Day

FROM

hospital_data

ORDER BY

Total_Stay_Day DESC;

--7. Total Patients Treated Per City

-- Count the total number of patients treated in each city.

```
SELECT
Location,
SUM(Patients_Count) AS Total_Patient
FROM hospital_data
GROUP BY Location
ORDER BY Total_Patient ;
```

--8. Average Length of Stay Per Department

-- Calculate the average number of days patients spend in each department.

```
SELECT
Department,
SUM(DATEDIFF(day,Admission_Date,Discharge_Date)*
Patients_Count) /
SUM(Patients_Count) AS Average_Days_Per_Department
FROM hospital_data
GROUP BY Department;
```

--9. Identify the Department with the Lowest Number of Patients

-- Find the department with the least number of patients.

```
SELECT
TOP 1
```

Department,

SUM(Patients_Count) AS Total_Patients

FROM hospital_data

GROUP BY Department

ORDER BY Total_Patients ASC;

--10. Monthly Medical Expenses Report

-- Group the data by month and calculate the total medical expenses for each month.

SELECT

DATENAME(month,Admission_Date) AS Month_Name,

SUM(Medical_Expenses) AS Total_Expens

FROM hospital_data

GROUP BY

DATENAME(month,Admission_Date),

MONTH(Admission_Date)

ORDER BY

MONTH(Admission_Date);